

6. What is the purpose of the numpy library? Write a program that creates a matrix of random integers between 1 and hundred having shape (3,4,4). [2+4]

Section C

Answer any 2 questions below a maximum of 1000 words, each carrying a maximum of 15 points. (2 X 15 = 30)

1. Answer the following questions:
 - a. How can you open a file in Python for both reading and writing from any place? Demonstrate it with a suitable program that allows reading writing from anywhere in the file. [3+5]
 - b. Write a function using NetworkX that can read an undirected graph from a given file whose each line contains edges as pairs of integers separated by a tab space. [7]
2. Answer the following questions:
 - a. What is Pandas? Discuss its key features and applications to machine learning. [2+5]
 - b. Create a class *PhDScholar* with attributes *registration number*, *registration date*, *enrollment number*, *name*, *date of birth* and *research title* and *supervisor name* . Your class should define a constructor to create an object of the class, a method to compute the age of the student, and a method to compute the time spent (in years) after registration from the current date. [8]
3. In an educational institution there are the courses of computer science having course code starting with CSC. Write a Prolog program for the following queries on the basis of the facts and rules given below: [7+2+3+3]

FACT: Following students studies the respective courses:

- I. Charlie studies the course csc135
- II. Olivia studies the course csc135
- III. Jack studies the course csc131
- IV. Arthur studies the course csc134

FACT: Following professors teaches the respective courses:

- V. Kirke teaches the course csc135
- VI. Collins teaches both course csc131 and course csc171
- VII. Juniper teaches course csc134

RULE: Consider the following rules

- I. X is a professor of Y if X teaches C and Y studies C

QUERY: Find the output for following queries:

- I. What does Charlie study?
- II. Who are the students of Prof. Kirke?